

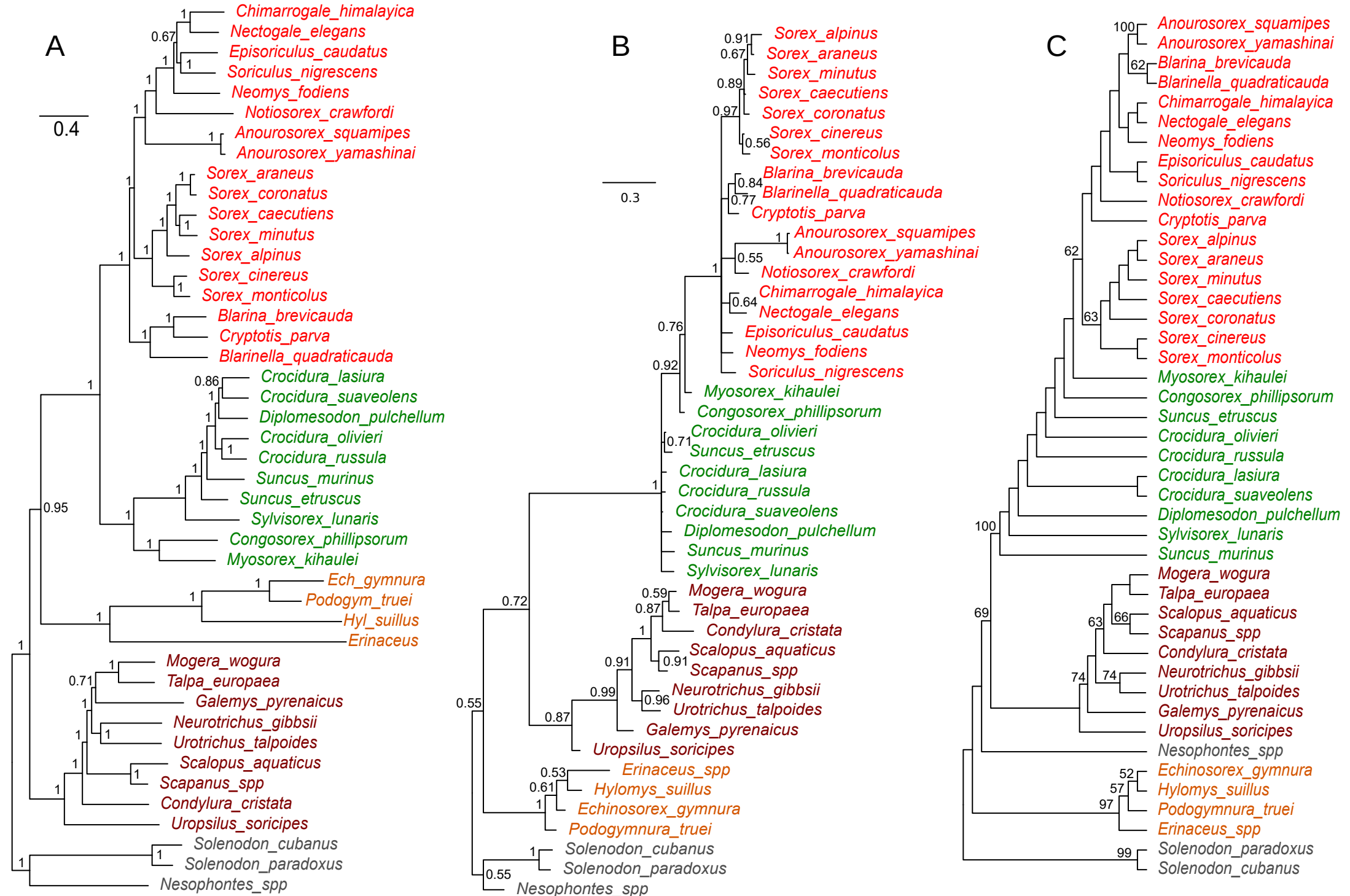


Figure S1. Optimal topologies of Recent taxa from non-timetree Bayesian analyses using A) DNA-indels and B) morphology, and C) parsimony with implied weights K=1 on morphology. Numbers adjacent to nodes represent posterior probabilities (A, B) and parsimony bootstraps, only reported when over 50 (C). Gray taxa are solenodontids, light brown erinaceids, brown talpids, pink heterosoricids, green crocidurines, red soricines.

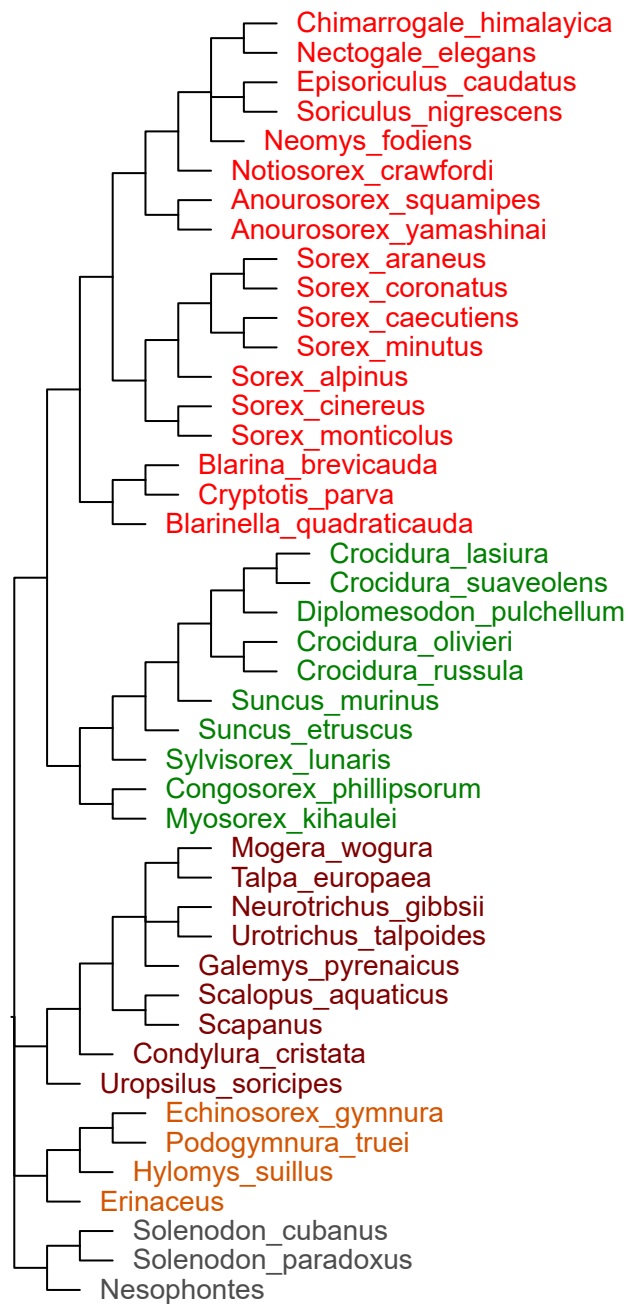


Figure S2. Well corroborated tree used to measure congruence in artificial extinction analyses. Topology is that derived from our Bayesian analysis of DNA-indels alone with nodes collapsed below a posterior probability of 0.8 (*Galemys*, *Neomys*) or when they conflict with recent, genomic analyses of Mammalia (lipotyphlan root, soricid sister taxon). Gray taxa are solenodontids, light brown erinaceids, brown talpids, green crocidurines, red soricines.

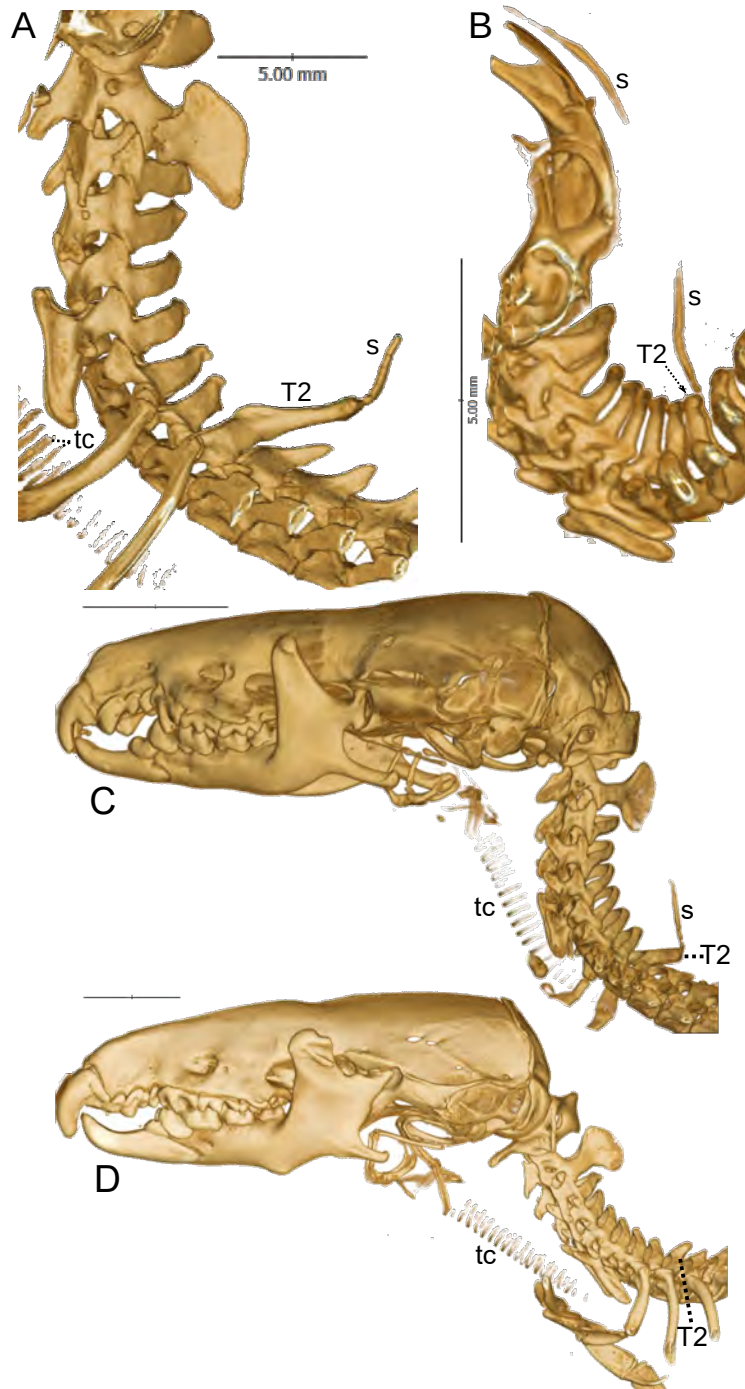


Figure S3. Lateral views of the neck region in alcohol-preserved, intact specimens of an erinaceid (A, *Hylomus suillus*, AMNH 278568) talpid (B, *Uropsilus soricipes*, AMNH 232373), and two soricids: *Crocidura russula* (C, UMMZ 157845), and *Blarina brevicauda* (D, UMMZ 154698). Note small nuchal sesamoids in A-C, lack thereof in D. Abbreviations are s = nuchal sesamoid, T2 = second thoracic vertebra, tc = tracheal cartilages. Scalebars = 5mm.

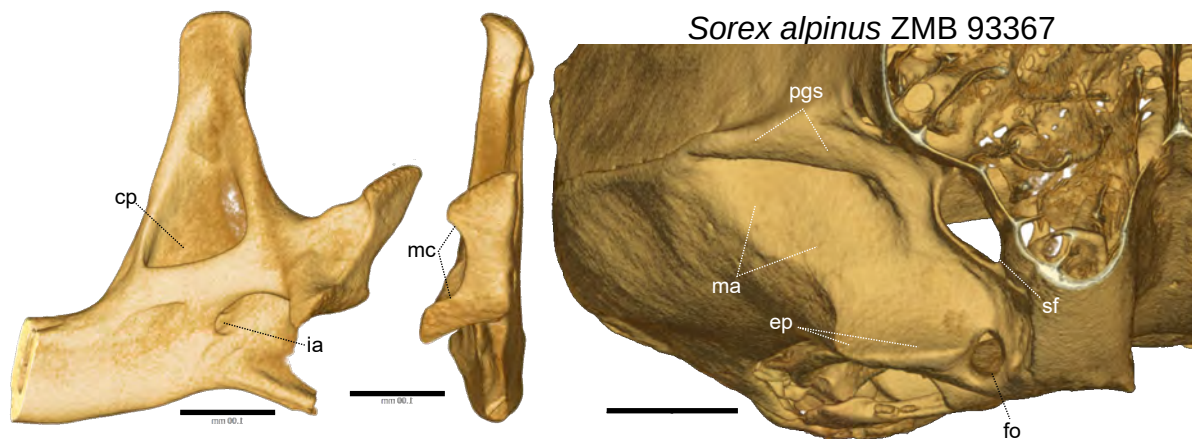
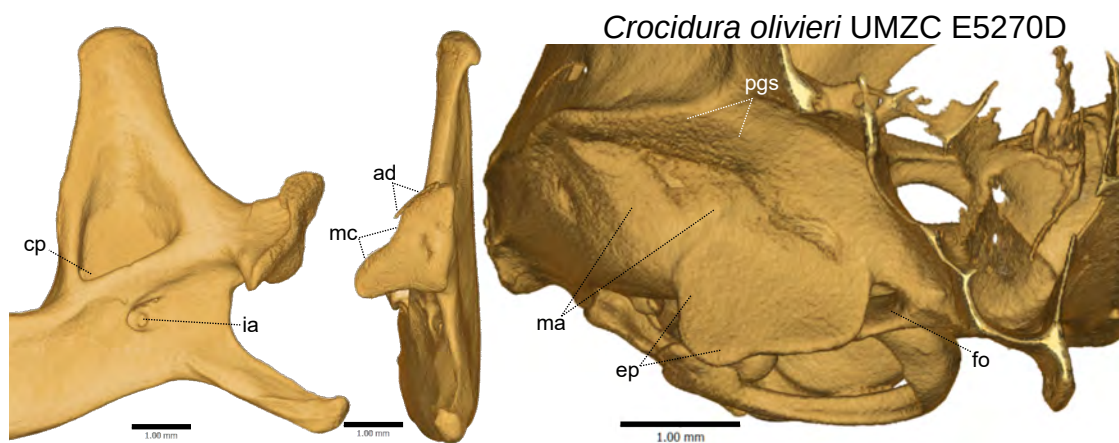
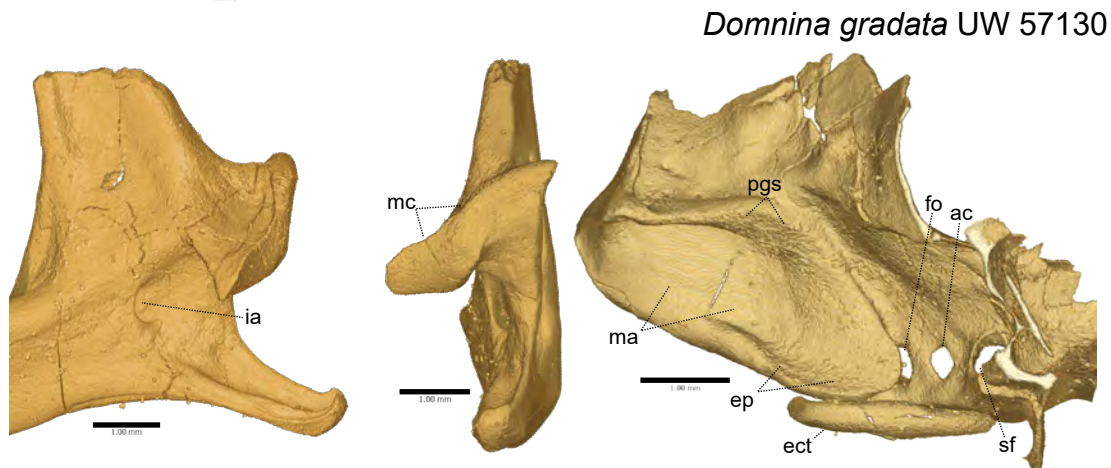
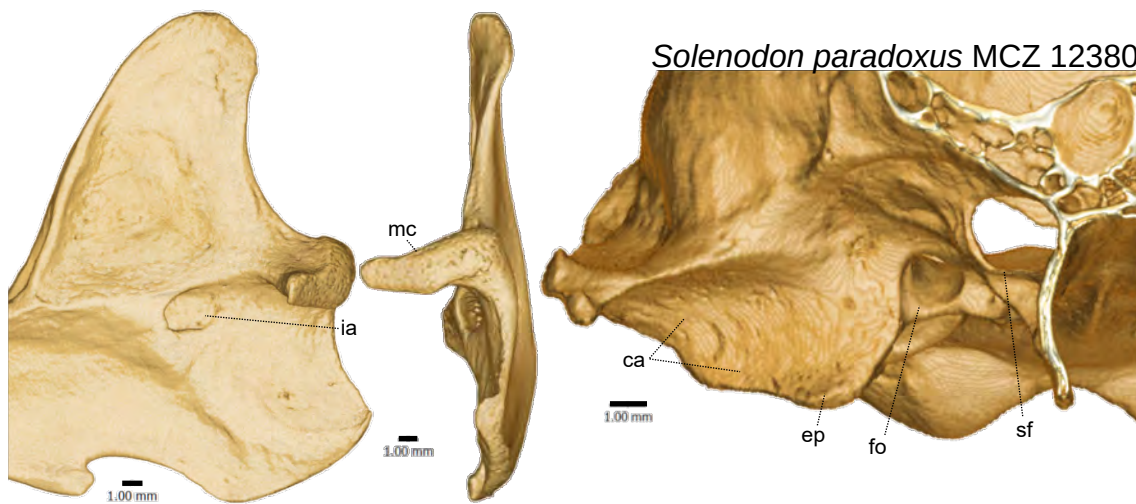


Figure S4. Right posterior jaws in lingual (left), posterior (middle) views, and closeup of squamosal glenoid fossa in anterior view, of (from top) *Solenodon paradoxus*, *Domnina gradata*, *Crocidura olivieri*, and *Sorex alpinus* (image reversed). ac = alisphenoid canal, ad = articular disk, ca = continuous articular area, cp = coronoid pocket, ep = entoglenoid process (= inferior glenoid process), ia = inferior alveolar foramen, ma = minimally-articular area, mc = mandibular condyle, pgs = superior glenoid process, sf = sphenorbital fissure. Scalebars = 1mm.

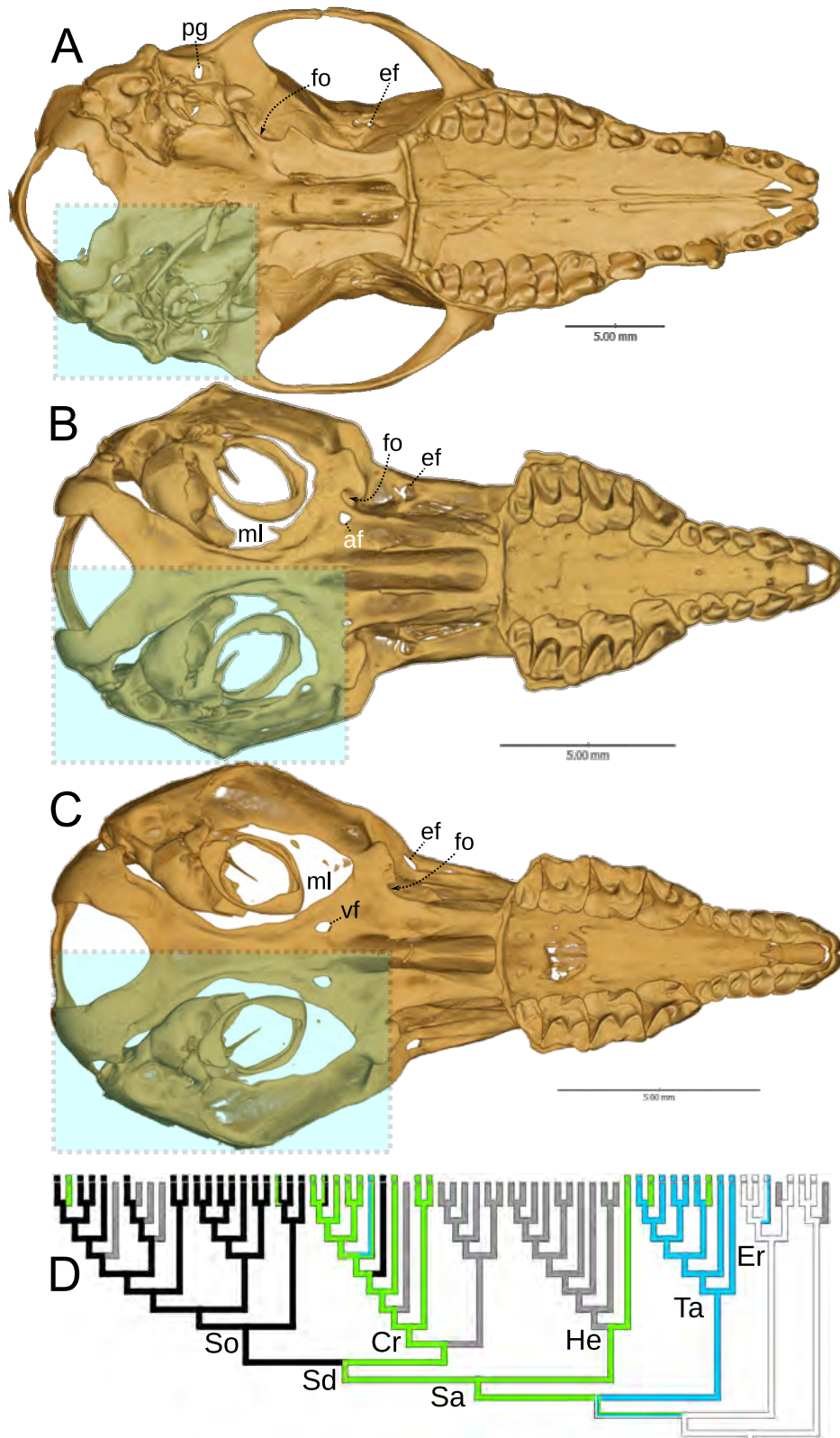


Figure S5. A) Proportion of the skull comprised of the basicranium in *Podogymnura truei* (A, FMNH 146591), *Sylvisorex lunaris* (B, FMNH 160192), and *Sorex coronatus* (C, ZMB 50225). Translucent blue rectangles are defined by foramen ovale anteriorly and occipital condyle posteriorly. Topology (D, taxa as in Fig. 4) shows scaled basicranial length (= foramen ovale to occipital condyle / condylobasal length): black > 41%, green 36-41%, blue 30-36%, white < 30%, gray = uncertain. Anatomical abbreviations are af = alisphenoid fenestra, ef = ethmoid foramen, fo = foramen ovale, ml = middle lacerate foramen, pg = postglenoid foramen, vf = vidian foramen. Taxonomic abbreviations are Cr = Crocidurinae, Er = Erinaceidae, He = Heterosoricidae, Sa = Soricoidea, Sd = Soricidae, So = Soricinae, Ta = Talpidae. Scale bars = 5mm.

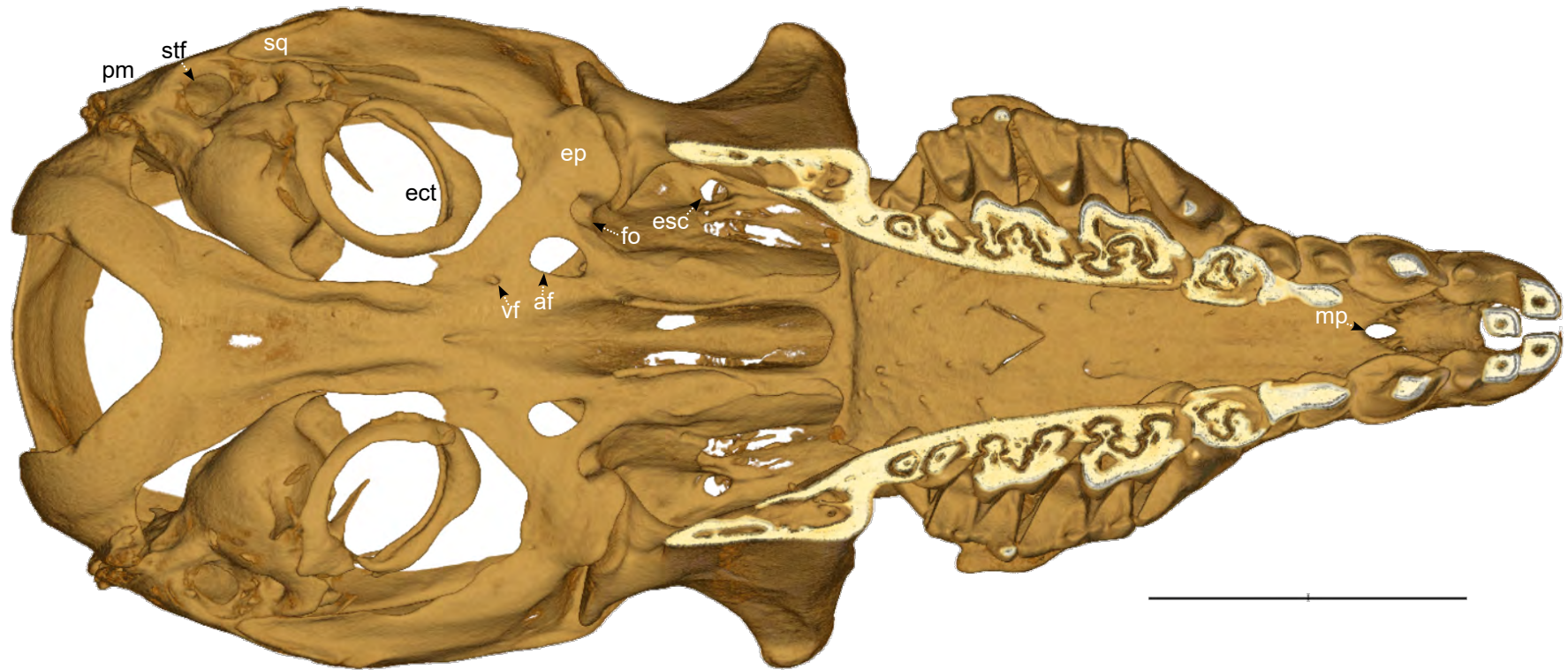


Figure S6. Ventral view of the skull of *Crocidura olivieri* UMZC E5270D. af = alisphenoid fenestra, ect = ectotympanic, ep = entoglenoid process, esc = ethmoid-sinus canal foramen, fo = foramen ovale, mp = median palatine foramen, pm = petromastoid, sq = squamosal, stf = gap in roof of stapedius fossa, vf = vidian foramen. Scale bar = 5mm.

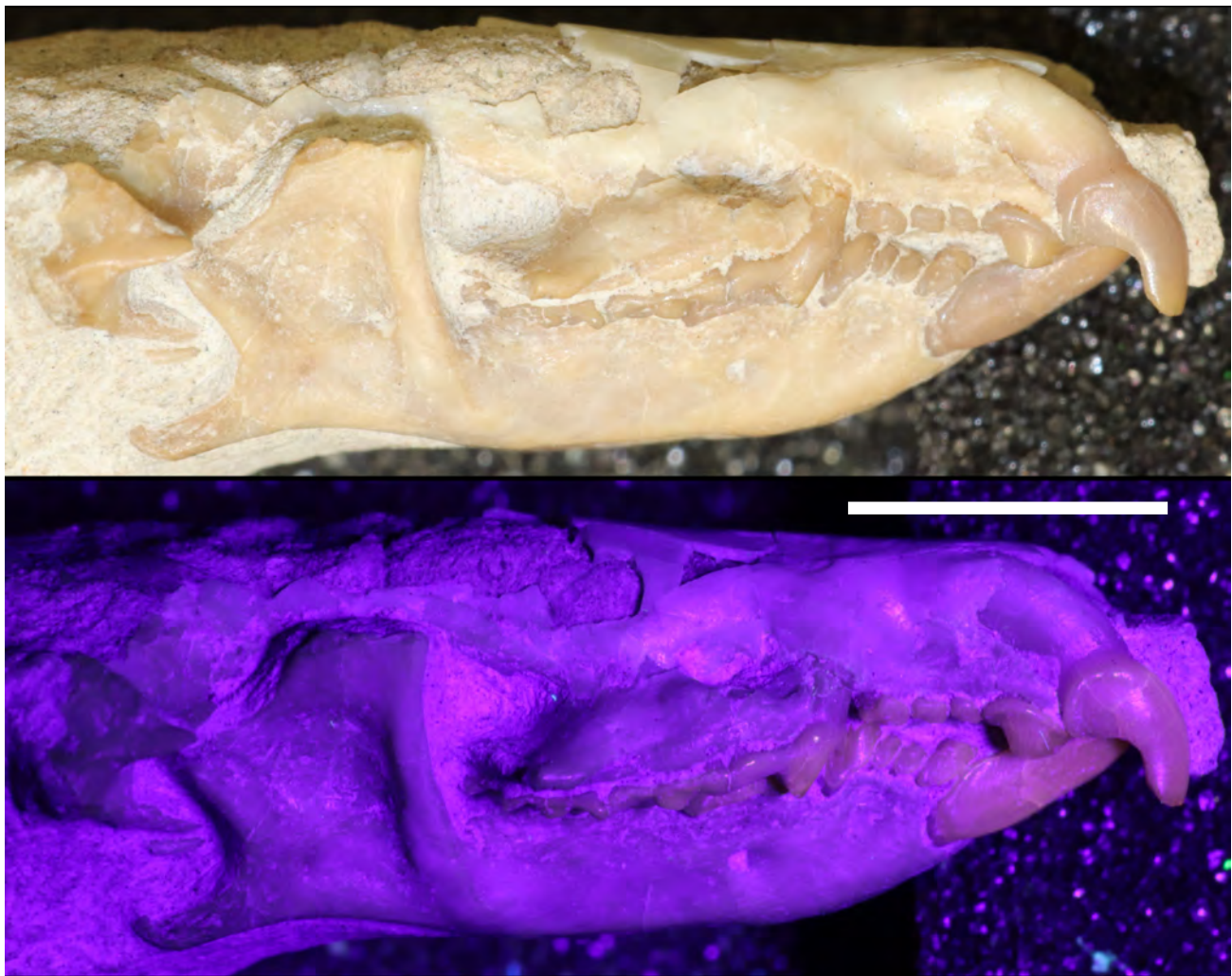


Figure S7. *Domnina gradata* (UW 57130) skull in lateral view photographed using visible (top) and ultraviolet (bottom) light. Note lack of obvious pigmentation and greater fluorescence of fossil matrix than dental tissues. Scalebar = 5mm.

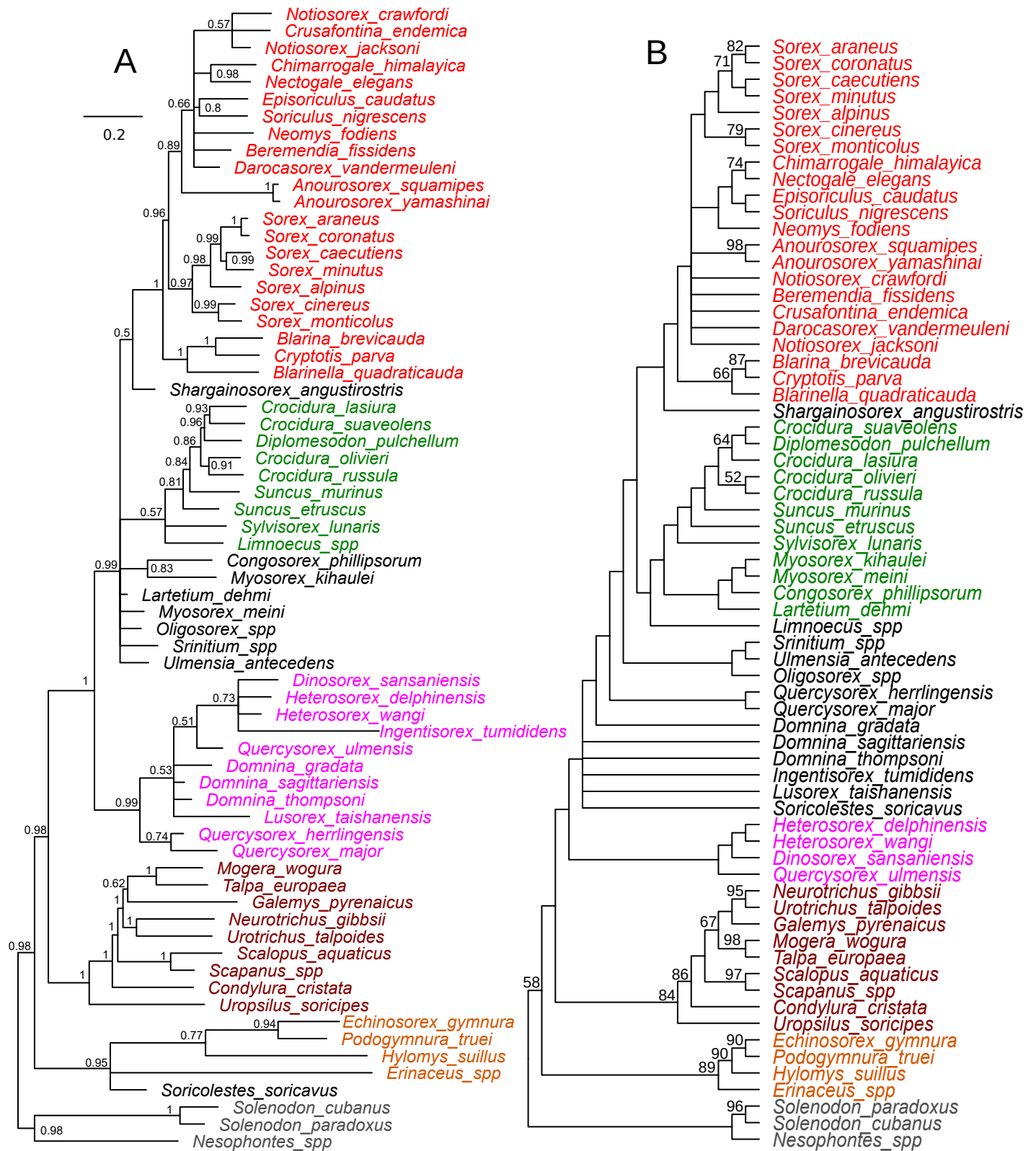


Figure S8. Optimal topologies from DNA-indel-morphology dataset derived from A) unconstrained, Bayesian majority-rule consensus of post-burnin topologies, discarding 25% of 40 million generations, sampling every 2000. B) Strict consensus of 54 shortest topologies, 5469.06771 steps, using parsimony implied weighting concavity value of 1. Parsimony branch lengths are arbitrary. Numbers adjacent to nodes indicate posterior probabilities (A) or bootstrap values, only given when over 50 (B). Gray taxa are solenodontids & *Nesophontes*, light brown erinaceids, brown talpids, pink heterosoricids, green crocidurines, red soricines.

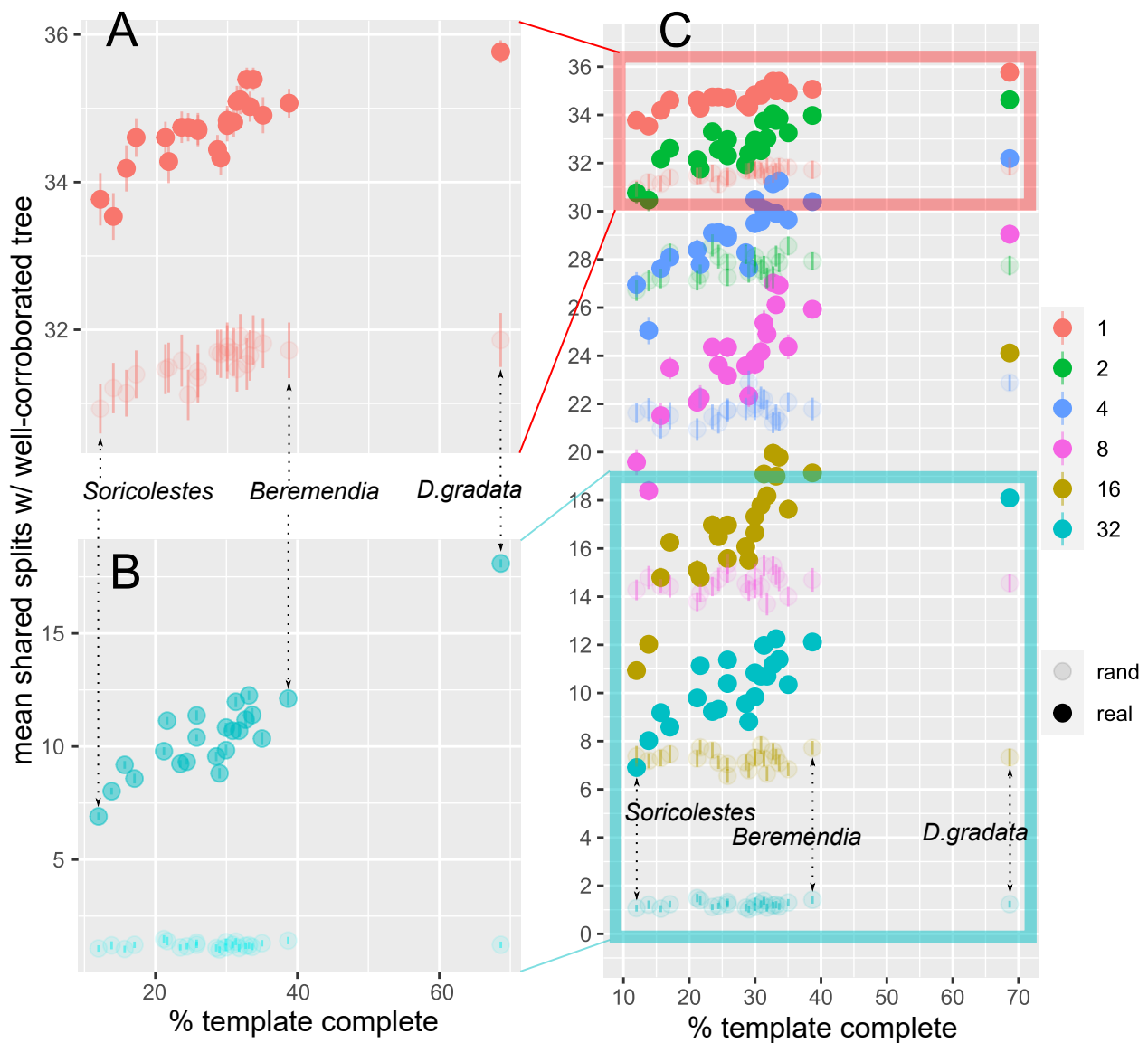


Figure S9. Results from artificial extinction experiments. In each graph, Y-axis represents mean shared splits with the well-corroborated tree; X-axis represents percent complete of a given fossil template (circles). Each circle is the mean of 43 artificially fossilized subjects for each template (e.g., 69% and 11% for *D.gradata* and *Soricolestes*, respectively, as indicated with arrows). Solid vertical lines extending from each circle represent $\pm$ one standard error from the mean ($\text{stdev}/\sqrt{43}$). Colors correspond to the number of subject taxa fossilized per analysis, from one (A), 32 (B) or all (C). Solid and translucent shades correspond to real vs. randomized character states, respectively.